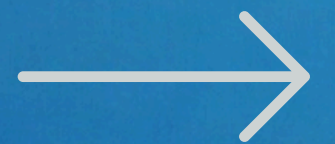


Analysis of Migraine Subjects' MMIDs



A brief Introduction



3rd

Most prevalent neurological disorder

2nd

Leading cause of disability (YLDs)

3:1

Women affected 3x than men

Migraine is a brain-based disorder, involving abnormal neuronal excitability and activation of the trigeminovascular system. Aura symptoms are explained by cortical spreading depression, while chronic migraine is associated with central and peripheral sensitization.

Source & Context

→ SOURCE

Vaghi G. et al. (2025)

Real-world effectiveness of anti-CGRP monoclonal antibodies over three consecutive one-year treatment cycles: An intention-to-treat analysis.

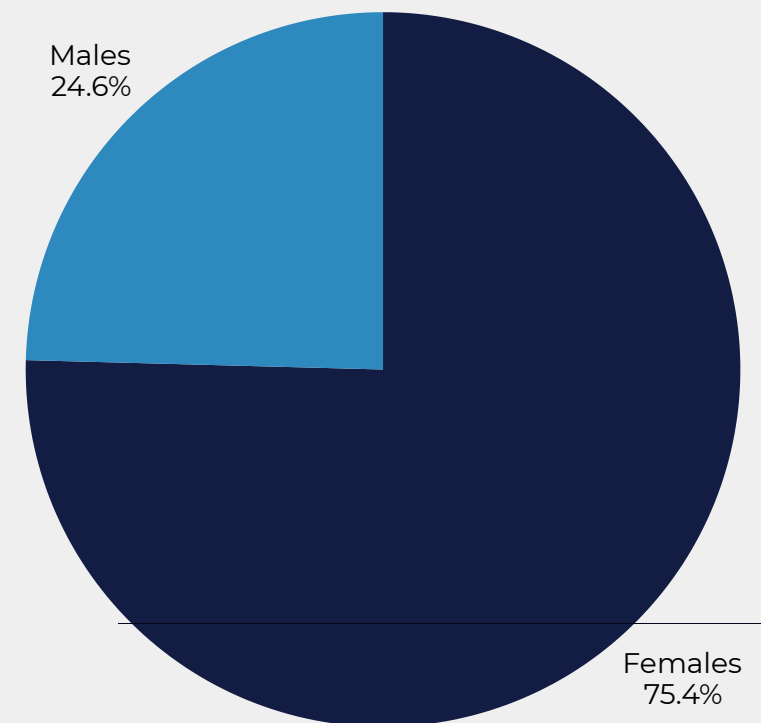
Cephalalgia, 45(8)

→ SETTING

An Italian multicenter study analyzing patients treated between 2018 and 2020.

→ Sample

179 patients



Mean Age: **51.3 years**

Study Design & Classification

→ **Prospective Cohort Study**

It follows a well-defined group of patients who initiated therapy within the same time frame

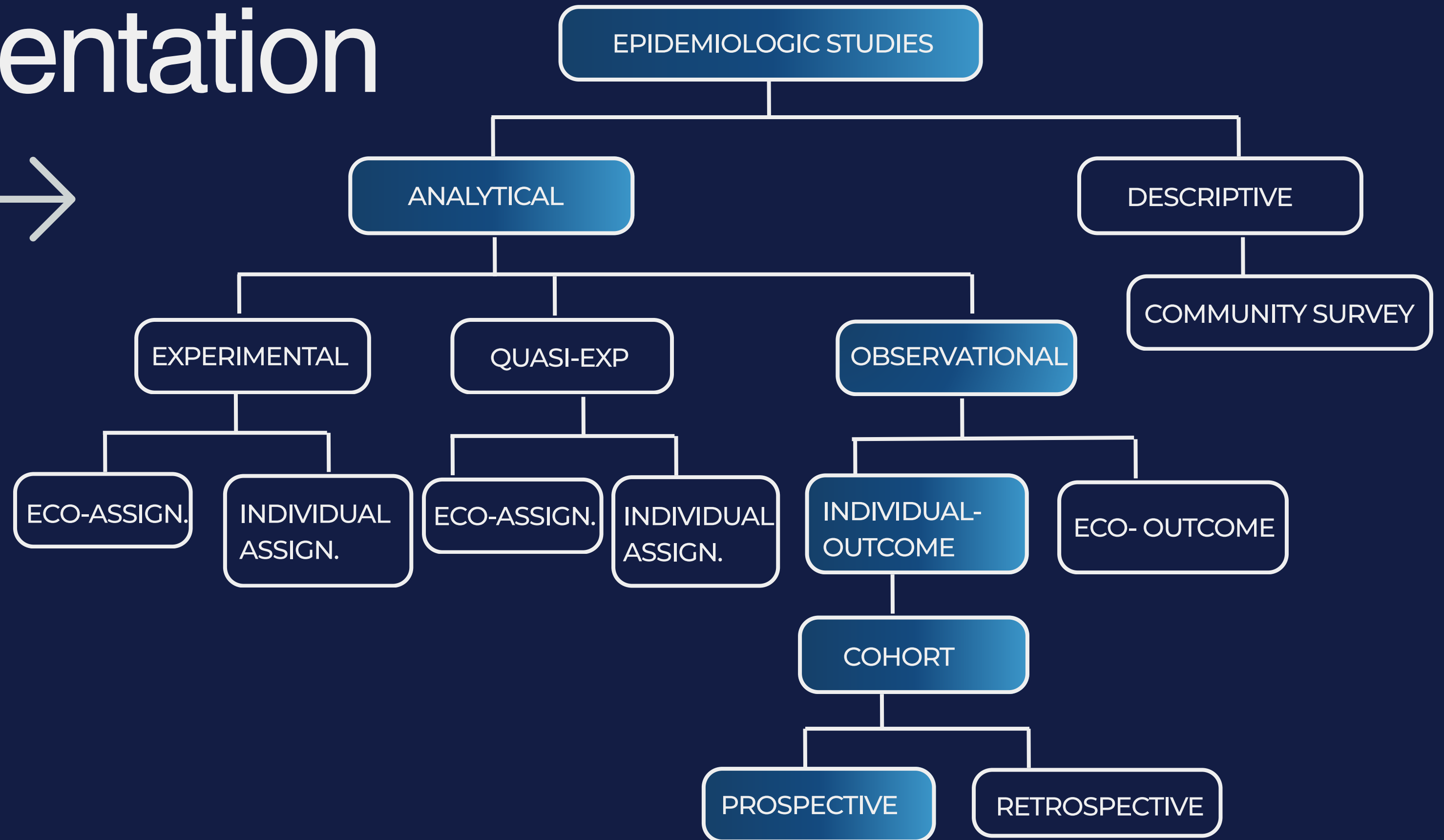
→ **Observational Real-world Study**

There is no randomization and no experimental manipulation of clinical conditions.

→ **Longitudinal Design**

Outcomes are assessed across three consecutive one-year cycles

Map Representation



The Intention-to-Treat (ITT) Approach

No Randomization

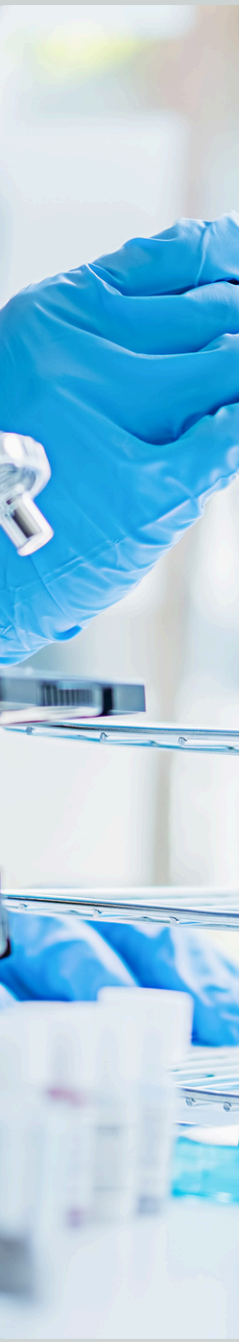
No placebo group or random assignment

Inclusion of All Subjects

Regardless of whether they completed the full three-year period or dropped out

Bias Prevention

To avoid "success bias"





Descriptive Analyses & Preprocessing

Dataset

Patient ID \ Features	Features	

Patient-Feature Matrix

- ✓ Non-narrative data
- ✓ Structured
- ? Selection Bias

Baseline: (179, 35)

Longitudinal: (2685, 14)

Baseline Data Exploration



Baseline Data Exploration

Baseline data exploration was conducted by grouping variables into clinically coherent domains to provide a structured and interpretable description of the study population.

Demographic features

Mean age:
~51 years

Gender:
Predominantly
female

Anthropometrics:
Moderate variability
(potential
confounders)

Disease History

Early onset
median 14 years

Long duration
median >30 years

High baseline
severity: median 28
MMDs, headache
intensity 8/10

Migraine Severity

Headache intensity:
median 8/10,

Weak correlation
MMDs ↔ intensity

>80% of patients in very
high severity category

Baseline Data Exploration

Baseline data exploration was conducted by grouping variables into clinically coherent domains to provide a structured and interpretable description of the study population.

Pain characteristics

Typical migraine features common

Aura in minority

Multiple pain profiles
(phenotypic heterogeneity)

Therapeutic refractoriness

Extensive prior acute & preventive treatments

High therapeutic refractoriness (median 4 failed drug classes)

Complex comorbidity profile
(psychopathology, sleep disorders)

Comorbidities & treatment discontinuation

50% discontinued therapy,

median treatment duration 23 months

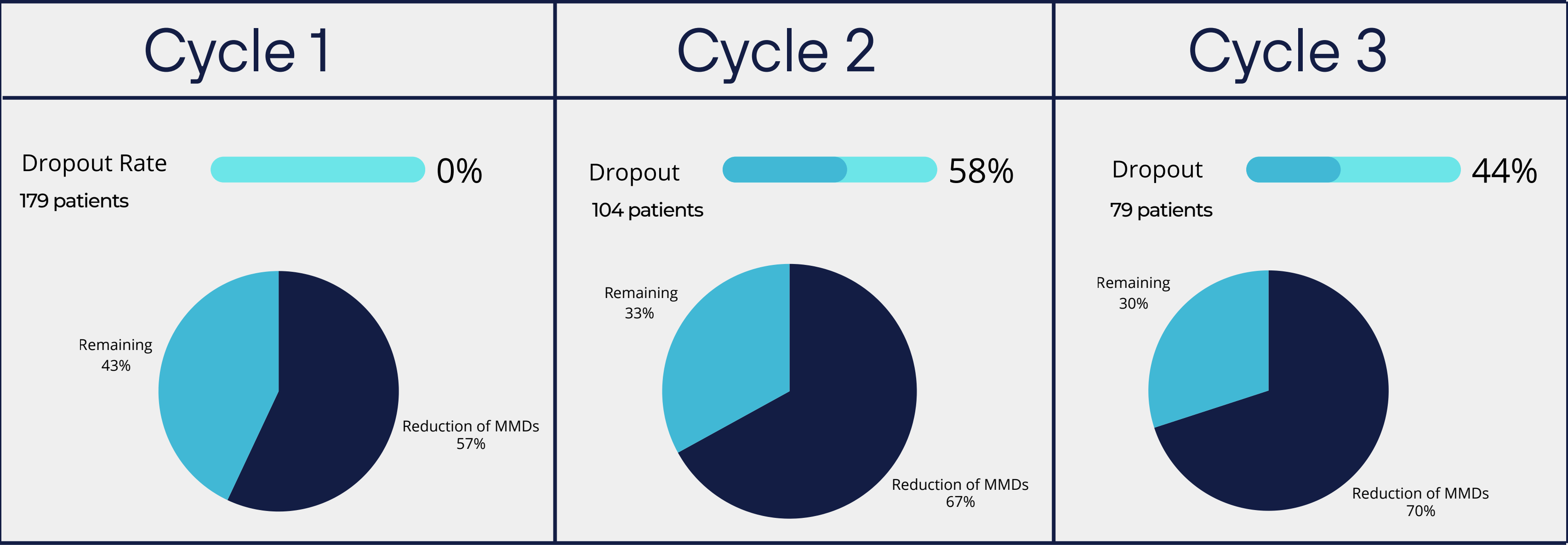
lack of a clear linear trend

Longitudinal Data Exploration

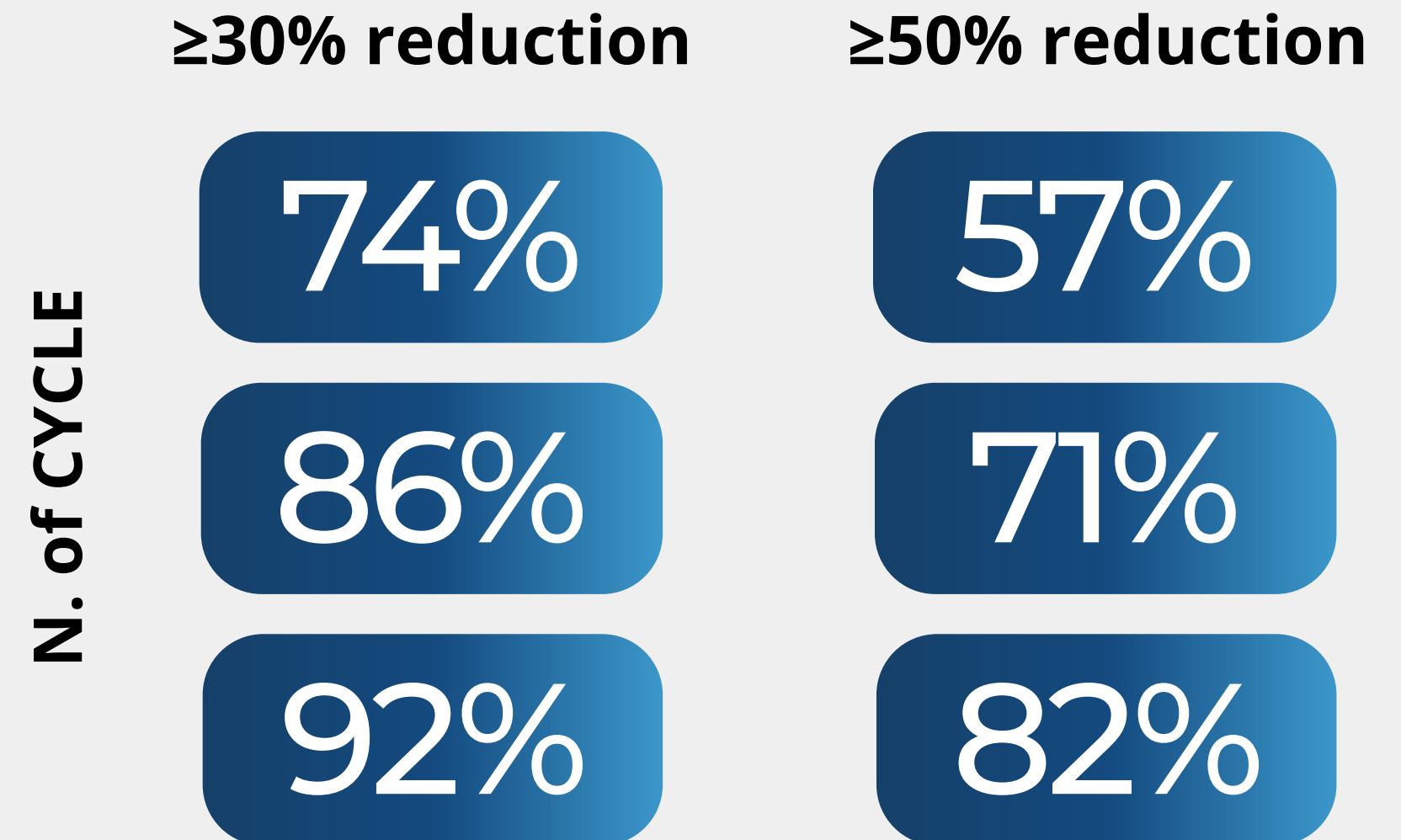
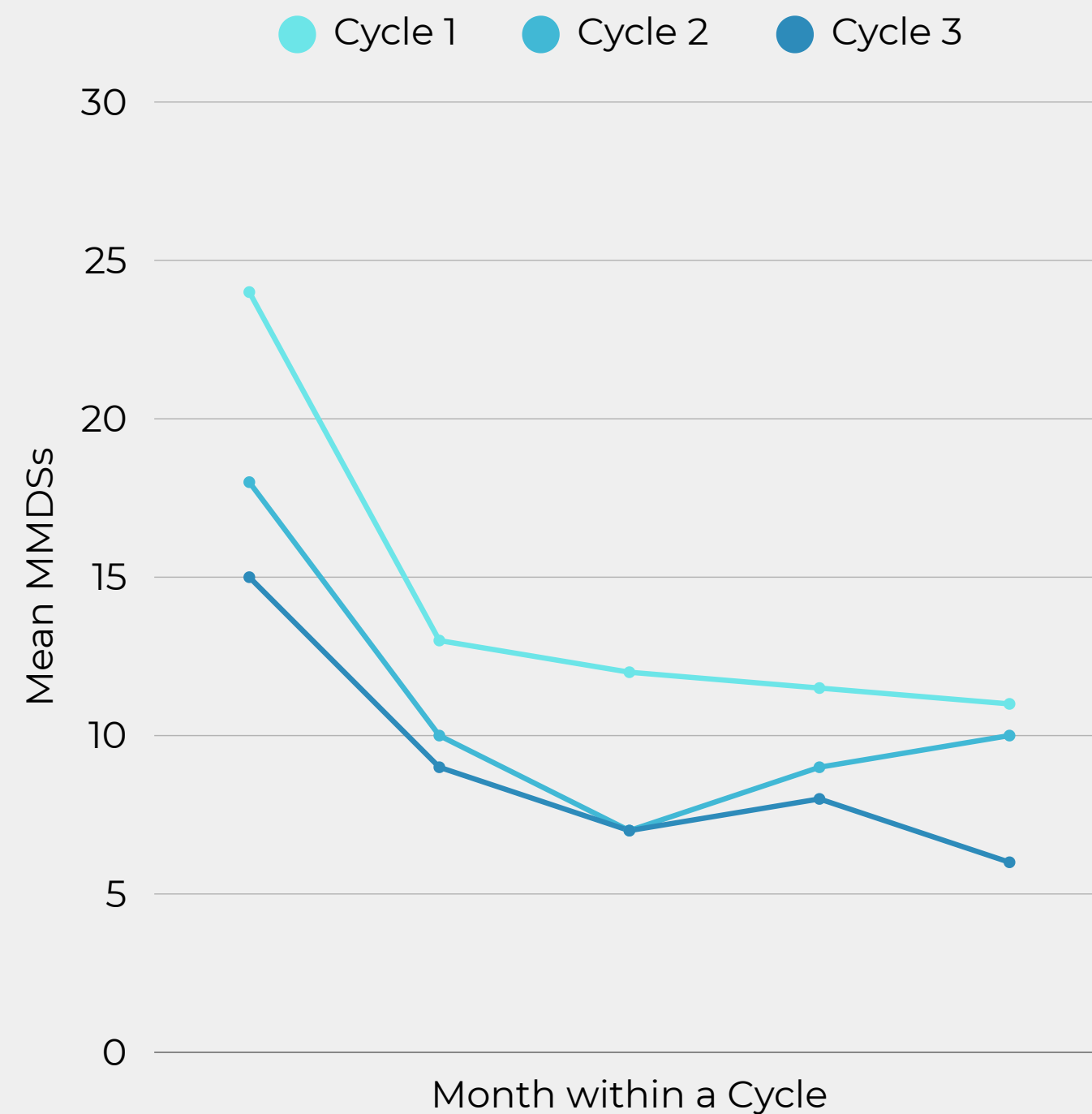


Longitudinal Data Exploration

The longitudinal dataset shows robust, sustained reductions in MMDs, consistent efficacy across cycles, and increasing responder rates, despite attrition and heterogeneous baseline characteristics.



Treatment Effect



These observations highlight a consistent treatment effect over time, with indications of maintained or enhanced benefit among continuing patients. The observed attrition suggests informative dropout, later-cycle should be interpreted cautiously.



Missing Data Imputation

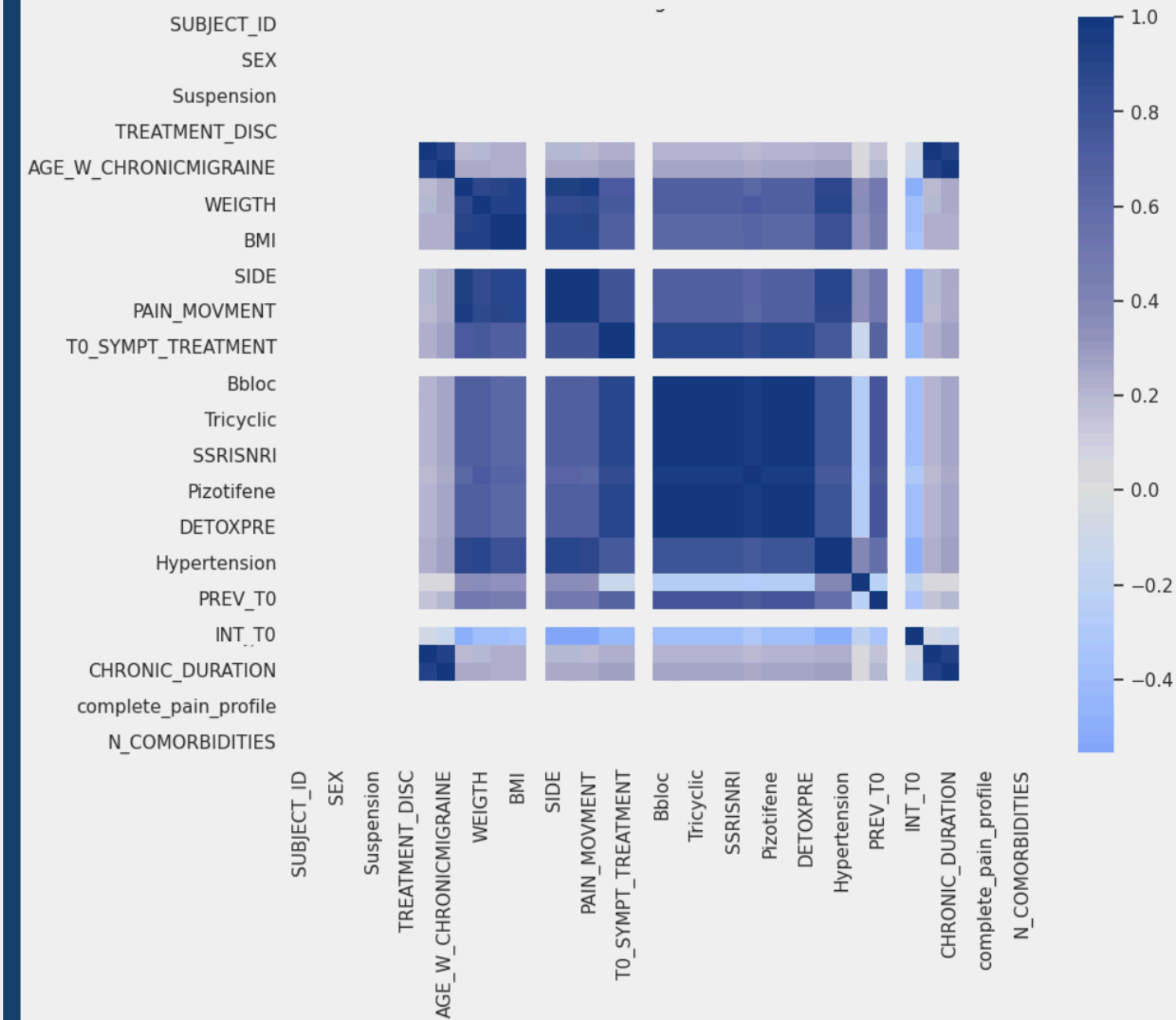
Baseline Imputation



Missingness Patterns

"Blocks" of missing data

MAR (Missing at Random) mechanism



Baseline Missingness Correlation

Workflow

Attempt 1:
**Complete-Case
Analysis (CC)**

Attempt 2:
**Simple Imputation
(Mean/Mode)**

Problem:

46/179 subjects retained

Reduced variability and ignored correlations.

Consequence:

Unstable model:
with ~40 predictors, the Condition
Number reached 1×10^{16}

It distorted the natural
distribution of clinical variables.



Relative RMSE of 0.35



It leverages Co-missing Clusters



Preserved full sample + realistic distributions.



Final Solution:

**MICE
(Multivariate Imputation
by Chained Equations)**

Longitudinal Imputation

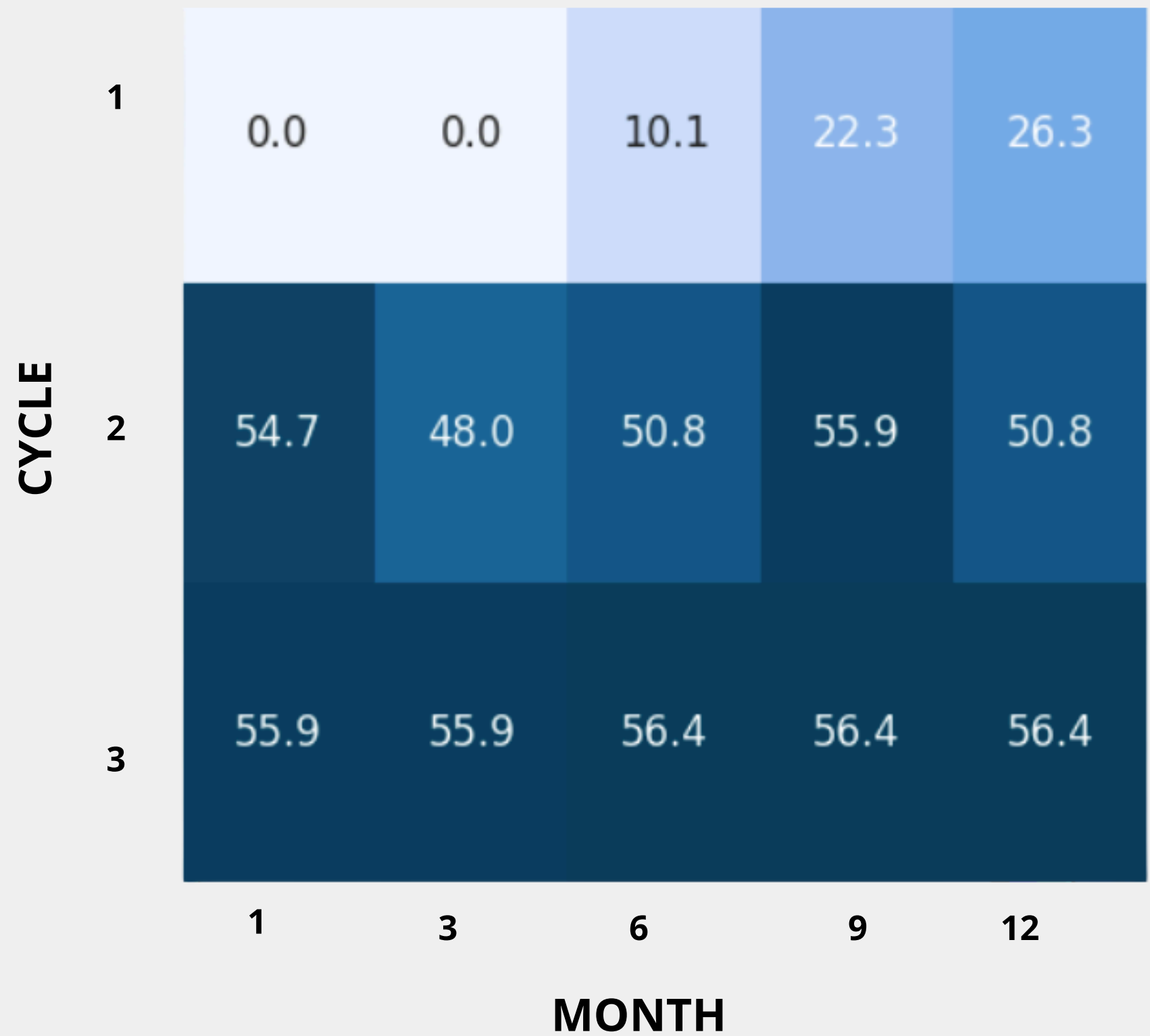


"Triangular" or monotone-like
increase in missingness

Progressive Dropout

Missingness Patterns

Missingness of MMDs by Cycle and Month (%)



Workflow

Attempt 1:
Linear Interpolation



0 imputed values

Attempt 2:
NOCB
(Next Observation Carried Backward)



28 values (6.5%) imputed

Result:



It recovered 217 missing values (50%)



The trajectories imputed provided a realistic Intention-to-Treat (ITT) dataset for all 179 subjects.



Final Solution:

LOCF
(Last Observation Carried Forward)



Predictors of Response and Discontinuation

Logistic Regression for Treatment Discontinuation



Outcomes – Predictors of treatment discontinuation

OR > 1	Baseline MMDs
	Number of Comorbidities

Increased risk of dropout

OR < 1	History of multiple failed treatments
------------------	--

Decreased risk of dropout

Mixed-Effect Model Analysis



Challenge

Nested Data Structure

Longitudinal Dataset

Multiple Observations per Patient

Observations from the same patient are **correlated**

Solution

Mixed Effect Model

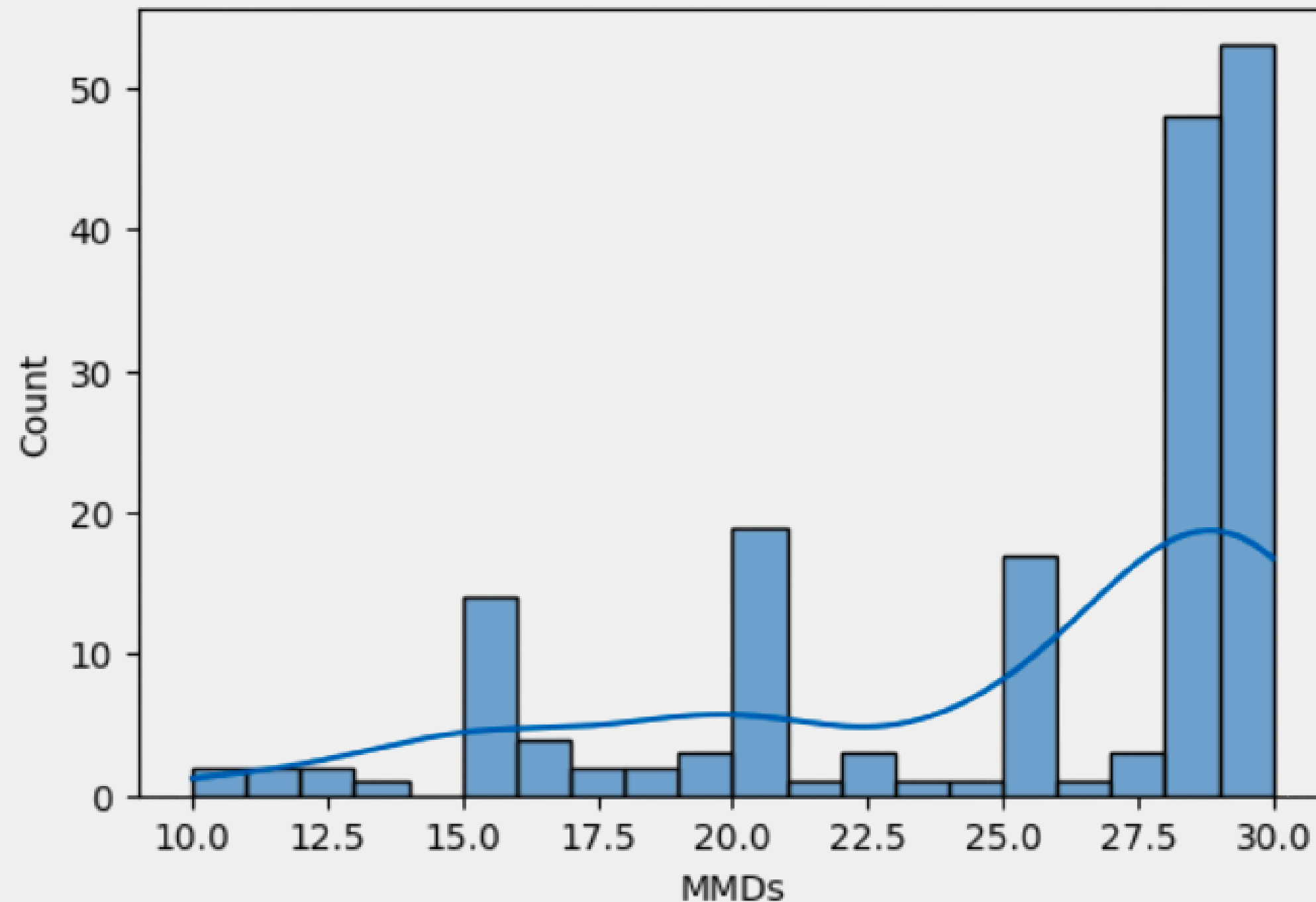
Fixed Effects

To measure the overall population trend

Random Effects

To allow each patient to have their own individual starting point

The graph shows that population starts with a wide spread of migraine frequency



Random
Intercept =
Variance

23.64

Confirms the significant differences in starting points between patients.

Results & Interpretation

Primary Outcome

-0.87 **MMDs per month**

Every month of treatment is associated with highly significant reduction in migraine days.

Outcomes

Baseline Predictors

-0.48 **Number of Failed Treatments**

More refractory patients showed a faster reduction in MMDs

0.55 **Initial Severity**

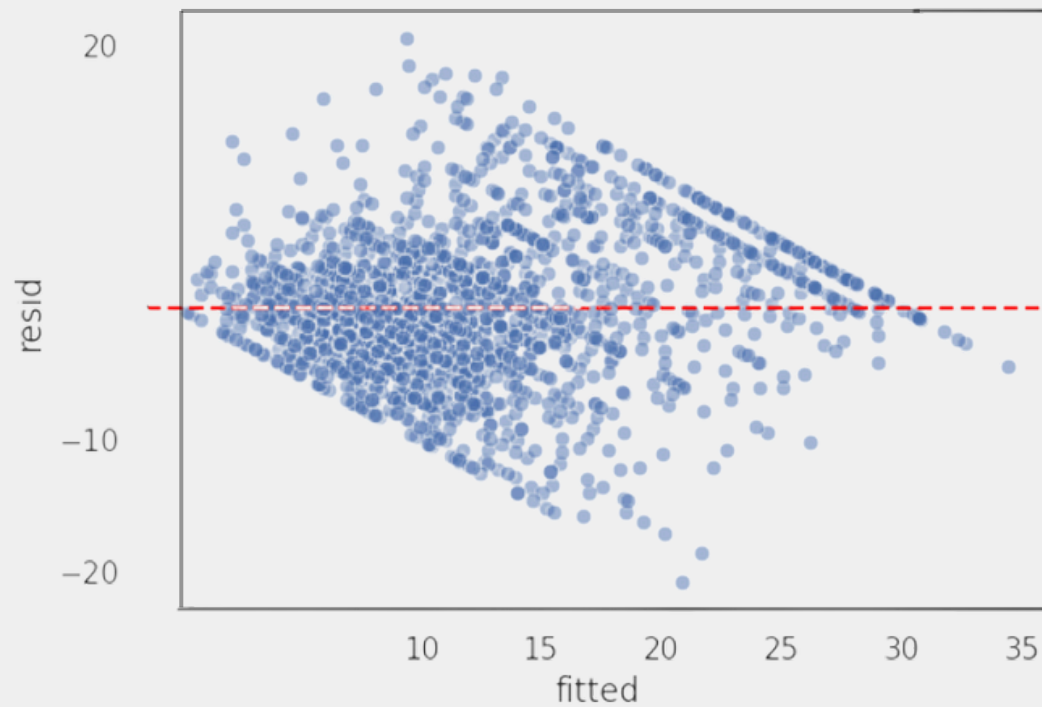
Starting frequency dictates the absolute level during follow-up

Demographics:

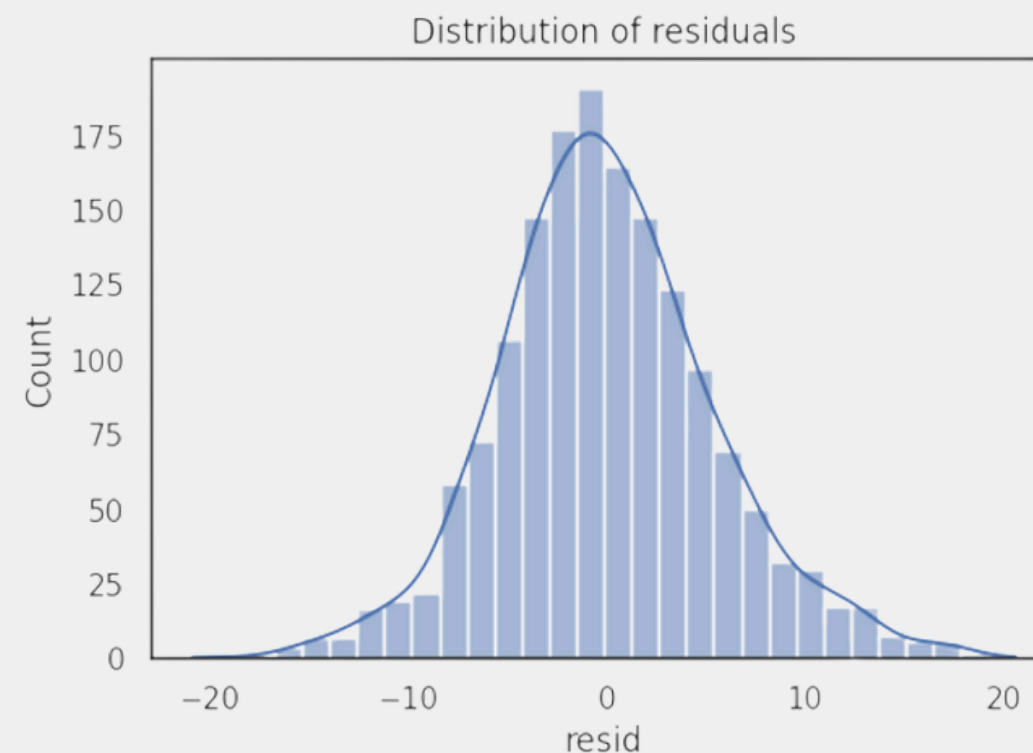
Age and **sex** were not significant, suggesting a consistent biological response across the cohort.

Model Validation & Robustness

Diagnostic Plots (Residual Analysis)



Plot shows no systematic error patterns, confirming the model's stability



Graph follows a Gaussian distribution, satisfying core LMM assumption.

Sensitivity Analysis

To address bias from dropouts, we re-ran the model on the **LOCF-imputed dataset**.

-0.73 **MMDs per month**

Robust and highly **significant**



Migraine Analysis

Presentation 2026

-

Electronic Phenotyping

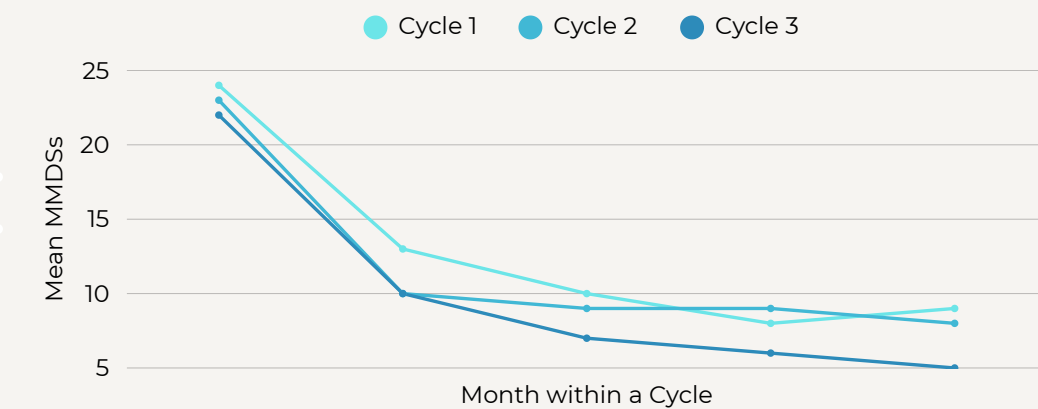
Cluster

Interpretation

Graphical Result

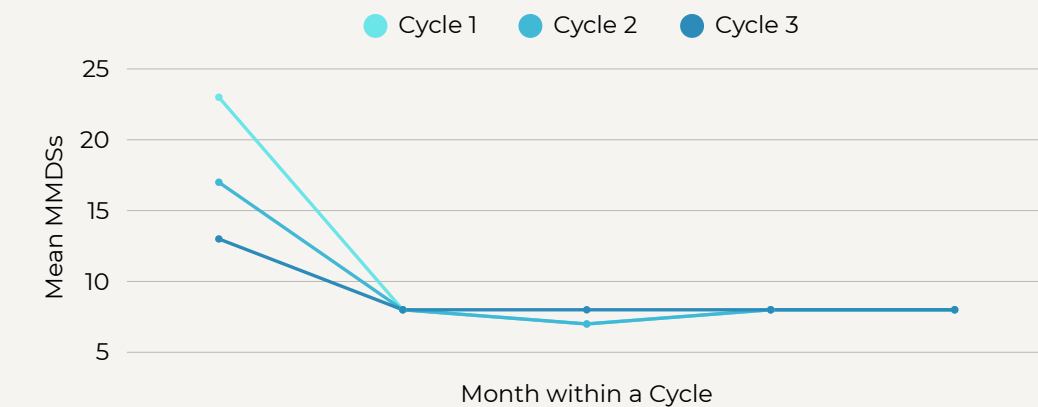
01: Stable Responder

Patients with a rapid, drastic reduction in MMDs early in Cycle 1, maintaining a consistently low burden (under 10 days) throughout the study



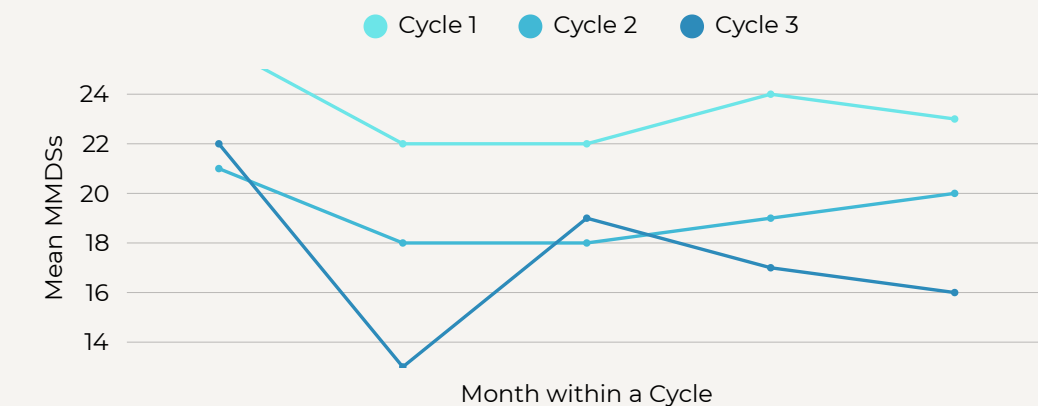
02: Partial Responders

Subjects who achieved significant reduction by the end of Cycle 1 and reached a stable, homogeneous state (approx. 7–8 days) in later cycles, despite a complex clinical history



03: Fluctuating

Characterized by high early volatility and failure to achieve stable reduction, remaining mostly above 15–20 MMDs



Final Consideration



Next Page

The analyses confirm that anti-CGRP therapies yield sustained and **substantial reductions** in monthly migraine days across multiple treatment cycles, while highlighting the critical role of patient heterogeneity.

By appropriately addressing missing data and accounting for within-patient variability, we showed that **treatment response** are driven primarily by **clinical history** and disease severity rather than demographic factors.

These findings underscore the value of translating longitudinal data into **actionable clinical insights**, ultimately supporting more informed and individualized decision-making in migraine care.

Thank You!